# Azure implementation

Which Azure resources, identities, and lifecycle rules run the reference deployment, and what changes for production.

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## DEPLOYED TOPOLOGY

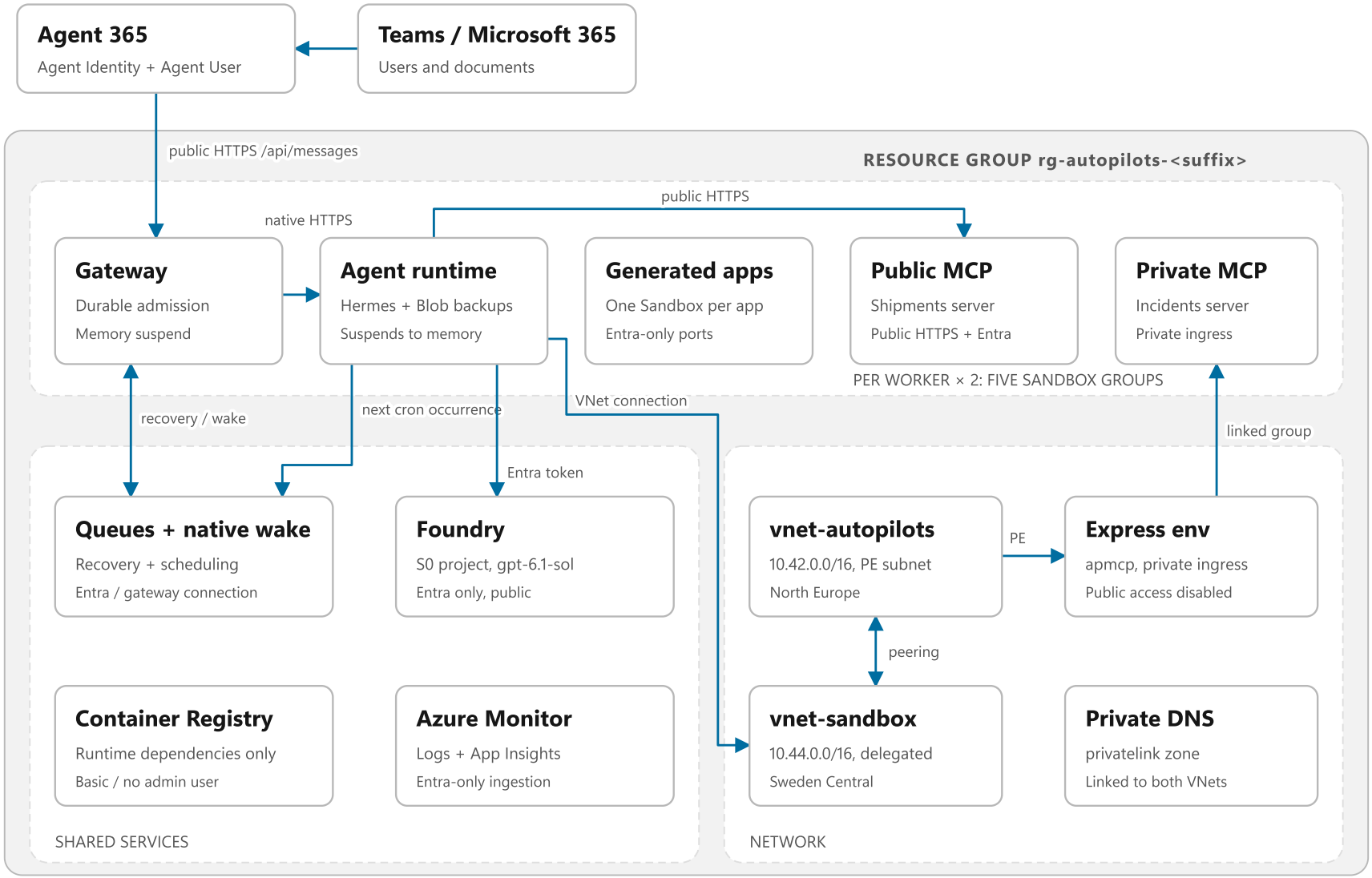
### Each Worker gets five Sandbox Groups; shared services stay in one resource group.

The reference deployment runs two Workers, `hermes` and `hermes2`, in one resource group. Sandbox Groups, Foundry, and monitoring run in Sweden Central; Service Bus, Container Registry, and the workload VNet run in North Europe.

The dependency base pins Hermes 0.22.6. Source releases, compute identity, and channel registration are independent. Each Worker also has a native managed receiver resource; it is not another always-on Sandbox.

Solid: connection in use  Dashed: optional connection or component

## 01 / Deployed Azure topology




Service Bus, Foundry, Container Registry, and Azure Monitor use Entra authentication on public endpoints. Only the private MCP path is private.

SOURCE

Platform layer · Worker layer · Networking

## IDENTITY AND ACCESS

01

### Separate send, receive, callback, and settlement authority.

Permanent roles are declarative; bootstrap write grants are temporary.

Each Sandbox Group keeps its user-assigned identity. A separate native receiver resource has a system identity. Azure integrations use Entra; existing operator and runtime-local API credentials remain application boundaries. No registry key, SAS connection string, or new application secret is introduced.

| PRINCIPAL | ROLE | SCOPE |
|---|---|---|
| Pinned deployer | Sandbox Data Owner | All five Worker groups; unchanged by operator login |
| Gateway identity | Sandbox Data Owner | Generated-apps Group only |
| Runtime identity | OpenAI User and AcrPull | Foundry account and runtime dependency registry |
| Gateway identity | Service Bus Data Sender | Recovery and enabled scheduling queue |
| Native receiver system identity | Data Receiver; Sandbox Data Owner | Worker queues; gateway callback group only |
| Gateway identity | Native connection named actions | Its own immutable-creator connection |
| Agent Identity | Service Bus Data Sender | Scheduling queue |
| Runtime and gateway identities | Monitoring Metrics Publisher | Application Insights |

Sandbox Data Owner permits more than execution, and the connection action wildcard permits more than completion. These are explicit preview-platform tradeoffs, not absolute least privilege. Deployment creates bounded bootstrap grants, verifies the caller-owned resource, and removes those grants in `finally`.

> SOURCE
>
> Worker role assignments · Monitoring roles

02

## Agent 365 objects come from scripts, not Terraform.

Blueprint, Agent Identity, and Agent User are Entra objects managed outside Terraform.

1. **Create identities**
   `worker_lifecycle bootstrap` creates the agent blueprint, Agent Identity, and licensed Agent User, and requests the delegated Microsoft 365 permissions.
2. **Federate the managed identities**
   `setup_identity` writes blueprint federated credentials for the gateway and runtime managed identities. It runs under a login that directly owns the blueprint.
3. **Register the endpoint**
   `setup_agent365` sets the messaging endpoint to the gateway URL plus `/api/messages` and publishes the Agent User package.

> SOURCE
>
> Deployment · Agent 365 setup · Token exchange

03

## Two Terraform layers; one workspace per Worker.

Infrastructure, images, and Sandboxes are separate, explicit steps.

PLATFORM

Resource group, registry, Service Bus namespace, Foundry, monitoring, VNets, Express environment, private endpoint, and DNS.

APPS

Per-Worker identities, groups, native receiver, queues, VNet connections, and permanent roles. Workspaces `autopilot-hermes` and `autopilot-hermes2`.

Build the runtime dependency image only when its pinned Hermes, installed packages, or compiled tools change. A source-only deployment packages the current files, verifies hashes, reuses prepared dependencies, and switches the supervised process in place. The deployment script creates caller-owned connections and fixed native receivers after services are healthy.

> SOURCE
>
> Deployment commands · Service deployment · Runtime deployment

04

## Registry conversion belongs to deployment.

Running workloads receive prepared disk-image IDs, not registry credentials.

**Terraform**
Creates infrastructure, identities, networking, and configuration.

**Build and convert**
Pins dependencies and converts each service image once.

**Deploy services**
Uses disk-image IDs, then records healthy endpoints.

Only the heavy runtime dependency image uses ACR native-v2 conversion and runtime-group `AcrPull`. Gateway and MCP services convert a public Python digest directly. No registry token or admin credential exists. Prepared group-scoped images

are reused.

The dependency base pins official Hermes 0.22.6 and contains LibreOffice, compiled .NET helpers, Node packages, and sealed-environment dependencies. Application source is not baked into that image. Thin-service requirements come from frozen locks; approved-feed wheels are downloaded during preparation and installed offline into a lock-addressed environment. A later source change reuses it without pip or image build.

> SOURCE
>
> Conversion identity decision · Conversion implementation · Commands and owner preflight

05

## Idle work needs no continuously running receiver.

Memory suspend preserves processes; source updates preserve compute identity.

| SERVICE | IDLE BEHAVIOR | UPDATE AND RECOVERY |
| --- | --- | --- |
| Gateway | Memory suspend after 300 s | Source switch preserves URL; journal restores from `gateway-state` |
| Runtime | 300-second Memory threshold; active-only self ingress | Source switch preserves compute; replacement restores native Hermes home |
| Private/public MCP | Memory suspend after 300 s | Prepared dependencies and source release; no custom ACR image |
| Generated apps | Independent five-minute suspend policy | Participant-authenticated child compute; default expiry 24 hours |

Before touching runtime compute, gateway maintenance rejects new admissions and checks unfinished work. Actual native task/executor counters protect the runtime source switch. Completed or quarantined receipts can await settlement without

repeating effects. A base-image or compute replacement may still change a native URL and require endpoint reconciliation. Superseded services remain tracked until the replacement passes real checks.

Native home archives and checked SQLite journal backups use platform-managed Blob volumes. SQLite snapshots and publication locks stay on local disk; only copied, renamed checkpoints reach Blob. Unknown channel effects are retained in the journal and dead-lettered, never reported as delivered. Deleting a Sandbox Group deletes its volumes.

Memory suspend is not disaster recovery, and stopped CPU/RAM is not a zero bill: Service Bus, ACR, storage, and monitoring retain independent charges. Managed source polling can delay scheduled work; it is not on the interactive Teams path. The rejected three-hop design took 161-175 seconds despite 4-13 seconds of model work. Normal chat now starts immediately and uses native Runs directly. Fast reply, scheduled-message settlement, and process retention are measured separately.

> SOURCE
>
> Service lifecycle · Runtime lifecycle · Memory suspend measurements · Generated apps

OBSERVABILITY

06

## Correlate application work without exporting its content.

W3C context crosses the bridge and wrapper; native callbacks expose model and tool execution.

Application Insights receives metadata-only OpenTelemetry spans. Worker and session attribution connect the gateway request, runtime invocation, model calls, and tools. Prompts, responses, bearer tokens, and tool payloads do not belong in this trace stream.

The Foundry external-agent registration links this execution to its monitoring surface. Registration does not move the agent into Foundry hosting or establish evaluation quality.

> Inspect correlation limits

The Sandbox-origin egress path preserves TraceId but can rewrite ParentId; its intermediate platform span is not available in application telemetry. Correlation is therefore useful without a complete platform-hop tree. The native `/v1/responses` path also lacks explicit session correlation.

An interrupted native handoff retains its lease to prevent misattributing later spans. Restart the affected wrapper/native runtime processes before reuse; do not fabricate a parent span or silently clear the fence.

SOURCE

Telemetry policy · Native handoff · Foundry registration

## PRODUCTION HARDENING

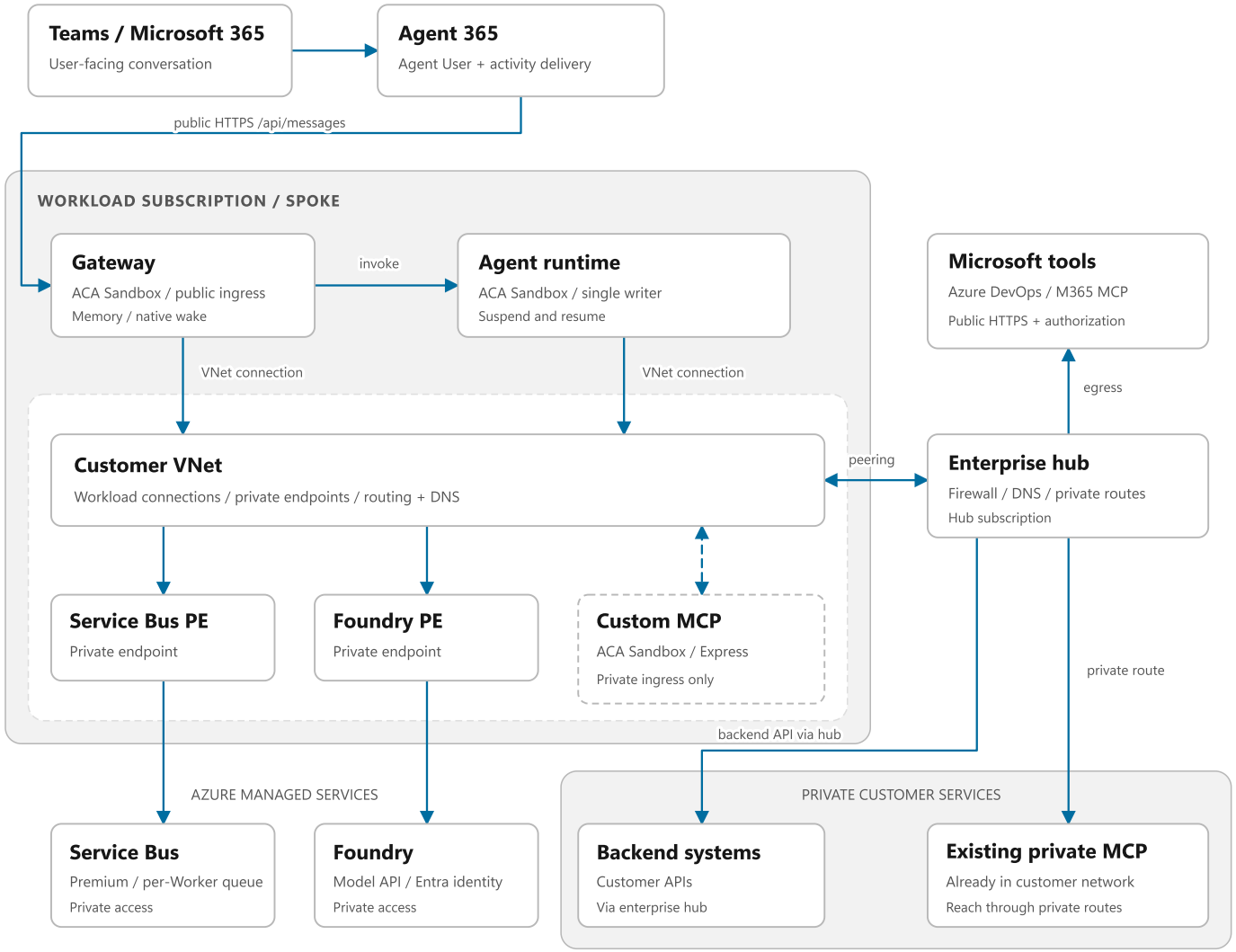
07

### Production adds private endpoints and an enterprise hub.

Network changes limit reachability; they do not replace identity checks.

This is a target design, not the deployed state. Service Bus moves to Premium, and Service Bus and Foundry get private endpoints in the workload VNet. A peered enterprise hub provides DNS, firewall, controlled egress, and routes to existing private MCP services and backend APIs.

## 02 / Production target network




PE means private endpoint. Custom MCP reaches backend systems through the workload VNet, peering, and enterprise hub before entering the private customer network.

Tools fit three patterns: Microsoft-hosted MCP services on public HTTPS, existing MCP services in the customer network, or a custom MCP adapter deployed with the solution. Identity checks stay the same in every pattern.

**Keep infrastructure declarative, source releases reproducible, native admissions durable, and every identity and recovery boundary explicit.**

### ▼ Agent 365

**Entra and Microsoft 365 objects**

Each Worker has an agent blueprint, an Agent Identity, and a licensed Agent User. Scripts create them and register the messaging endpoint as the gateway URL plus `/api/messages`. Terraform does not manage these objects.

### ▼ Teams and Microsoft 365

**User-facing workloads**

Users mention the Agent User in Teams, mail it, or comment in Office documents. Work IQ and Graph calls from the runtime run as the Agent User.

### ▼ Gateway Sandbox Group

**Memory suspend after 300 s**

The bridge validates Agent 365 Activity Protocol authentication and handles Teams turns directly through native Runs over OnDemand HTTPS. Interactive messages are not persisted to a recovery queue. Its managed identity owns the scheduling connection and generated-app lifecycle; it cannot provision runtime compute. The journal is checkpointed to the `gateway-state` Blob volume for scheduling receipts and bounded conversation state.

### ▼ Agent runtime Sandbox Group

**Hermes 0.22.6 / 300-second Memory threshold**

A dependency image and separately hashed source release host one native writer. Model requests use native Runs over authenticated OnDemand HTTPS; there is no runtime queue consumer or custom transport journal. Native home archives use the `hermes-backups` Blob volume. Actual native task/executor counters protect updates. Its VNet connection reaches private MCP; source changes need not replace compute.

### ▼ Generated apps Sandbox Group

**One Sandbox per app**

The gateway starts each Worker-created app in its own Sandbox with Entra-protected ports. Apps suspend after 5 idle minutes and expire after a lifetime of 24 hours by default (15 minutes to 7 days).

### ▼ Public shipments MCP

**Public HTTPS / Memory after 300 s**

A public Python base plus verified source release serves demo shipments. Entra enforces tenant, issuer, audience, and approved shipment scope/application role. The native port is OnDemand; no registry grant is needed.

### ▼ Private incidents MCP

**Private ingress / Memory after 300 s**

The group remains irreversibly linked to the Express environment. Application authorization pins the expected Worker Agent Identity and incident role. Its public Python base and source release need no ACR image.

### ▼ Service Bus and native receivers

**Standard / Entra / scheduling queue only**

Each Worker has one queue for enabled cron, Dreaming, and deterministic publication retries, with a fourteen-day TTL. No activity-recovery queue exists. Locks are five minutes; duplicate detection is one hour. The native receiver wakes only the gateway. The gateway renews and settles scheduled deliveries through its creator-bound connection, using durable receipts and bounded active-work keepalive.

### ▼ Foundry

**AIServices S0, local auth disabled**

Project `autopilots-project` with deployment `gpt-6-1-sol` (model `gpt-6.1-sol`, GlobalStandard, capacity 100) in Sweden Central. The runtime managed identity has Cognitive Services OpenAI User. Foundry also holds the external-agent registration and an Application Insights connection.

### ▼ Container Registry

**Runtime dependency base only**

Basic registry with admin disabled. It holds the pinned Hermes dependency image, compiled Office tools, and required runtime packages. Only runtime conversion needs AcrPull. Gateway and MCP source changes use public-image releases and never need a custom image build.

### ▼ Azure Monitor

**Log Analytics + Application Insights**

Both have local authentication disabled. Gateway and runtime managed identities publish metadata-only OpenTelemetry spans with Monitoring Metrics Publisher.

### ▼ Workload VNet

**10.42.0.0/16, North Europe**

Holds subnet `snet-private-endpoints` (10.42.2.0/24) with the private endpoint `pe-apmcp-*` for the Express environment.

### ▼ Express managed environment

**Public network access disabled**

Environment `apmcp-*` in Sweden Central supplies private ingress for the linked private-MCP Sandbox Group.

### ▼ Sandbox VNet

**10.44.0.0/16, Sweden Central**

Subnet `snet-sandbox` (10.44.0.0/24) is delegated to `Microsoft.App/environments`. Runtime and gateway Groups attach through VNet connections. Peered in both directions with the workload VNet.

### ▼ Private DNS zone

**privatelink.swedencentral.azurecontainerapps.io**

Resolves the private MCP hostname to the private endpoint. Linked to both VNets.

### ▼ Teams and Microsoft 365 entry points

**Reference implementation**

Personal messages and explicit Agent User mentions are supported entry points. Delivered replies, selected reactions, Mail notifications, and Office comment events have workload-specific handling.

Shared groups are a shared audience. Private per-person delivery and human on-behalf-of identity are not implemented.

### ▼ Agent 365 channel and identity binding

**Channel identity**

Each Worker has an Agent Identity and Agent User. Agent 365 binds its endpoint and permissions, then delivers Microsoft 365 activities to `/api/messages`. The Agents SDK authenticates incoming requests.

### ▼ Gateway Sandbox

**Memory suspend / immediate native Runs**

Authenticated Teams activities use the direct bridge-to-runtime path. The gateway does not persist interactive payloads or create recovery references. The journal retains schedule receipts and bounded conversation state with a checked Blob backup. Model inference and tool calls remain in the runtime.

### ▼ Agent runtime

**ACA Sandbox / native Runs**

Deployment owns compute. The runtime owns execution of native Runs. Active-only keepalive protects work. Memory resume preserves processes; source releases preserve the Sandbox ID and URL. Native home archives recover runtime state, while the independent gateway journal preserves scheduling receipts and bounded conversation state.

### ▼ Customer VNet

**Workload connectivity**

Gateway and runtime attach here. Private endpoints connect managed services; private MCP ingress joins the same network boundary. Peering reaches the enterprise hub. Routes and DNS provide reachability; identity still controls access.

### ▼ Service Bus private endpoint

**Target design / receiver routing must be verified**

Private endpoint and DNS can connect workload clients to Premium. A managed receiver runs outside the actor, so actor VNet attachment alone does not establish receiver reachability. Verify the supported native connector network path before disabling public broker access. Entra authorization remains required.

### ▼ Foundry private endpoint

**Target design**

The Foundry private endpoint connects model access to the workload VNet. Private DNS resolves the model endpoint to its private address; the runtime still authenticates with its managed identity.

### ▼ Custom MCP

**Optional adapter**

Add an ACA-hosted adapter when a backend lacks MCP. It exposes a private MCP interface and reaches backend APIs through the workload VNet and enterprise hub. Use a supported private-ingress hosting configuration.

### ▼ Service Bus queue and dead-letter handling

**Target: private broker plus verified receiver route**

The single per-Worker queue carries scheduled metadata only; it is not an activity-recovery or request/reply transport. Native receivers own PeekLock; the gateway renews and settles through its own connection. A private endpoint is not proof of receiver reachability. Source polling affects scheduled work only. Unknown effects stop, and DLQ is visible failure state, never successful delivery.

### ▼ Foundry model inference

**Target design: private endpoint**

The agent runtime calls Foundry through the private endpoint. Runtime managed identity and RBAC authorize inference; private routing and DNS provide reachability. The model service remains outside the workload subnet.

### ▼ Enterprise hub

**Shared network services**

VNet peering connects the workload spoke to hub routing, DNS, and firewall services. The hub provides controlled public egress and private routes to existing customer services, including MCP endpoints.

### ▼ Microsoft-hosted MCP tools

**Public HTTPS**

Use approved Microsoft-hosted tools, such as Azure DevOps and Microsoft 365 MCP integrations. Configure permissions for each service. Microsoft 365 operations use the approved Agent User flow; other services require their own supported connection.

### ▼ Existing private MCP

**Customer-provided service**

Connect to an MCP server already running in the customer network through private routing and DNS. Reuse its approved authentication and permissions; no new MCP host is needed in this solution.

### ▼ Backend systems

**Customer APIs**

Custom MCP reaches these APIs through the workload VNet, peering, and enterprise hub into the private customer network. Backends keep their existing interfaces and permissions; they do not need to implement MCP.